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Symbol sequences and orbits of the free-fall three-body problem

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Abstract

Using the symbols and symbol sequences along the orbits introduced in our preceding work, we numerically study the orbital structure of the free-fall three-body problem. We confirm and re-interpret the results obtained by us before. We describe the overall structure of the plane. It turns out that the structures of the initial condition plane can be systematically obtained with symbol sequences. Then, we obtain the structure of two interesting local regions: the isosceles and collinear boundaries of the plane. We present sequences of triple collision orbits and periodic orbits on these boundaries. We additionally argue that stable and/or unstable manifolds of the two-body collision manifolds connect different triple collision manifolds.

Key words: celestial mechanics — chaos — three-body problem

1 Introduction

We are interested in the structure of the initial condition plane of the free-fall (i.e., zero initial velocities) three-body problem (Agekyan & Anosova 1967; Tanikawa et al. 1995; Tanikawa & Mikkola 2008).

Montgomery (1998) introduced symbolic dynamics in his variational search for periodic orbits in the planar three-body problem. He proved the existence of periodic orbits. Moeckel, Montgomery, and Venturelli (2012) proved through the variational method the existence of free-fall periodic orbits (i.e., "brake" periodic orbits in their terminology) passing through collinear syzygy twice during one period in the isosceles free-fall three-body problem.

We independently started the study of the planar three-body problem including the free-fall cases using symbol sequences, in which the structure change of the initial condition plane according to the value of the angular momentum was the target (Tanikawa & Mikkola 2008). In the present paper, we study purely numerically the structure of the initial condition plane of the free-fall problem. We consider the equal mass cases so that we only need to treat 1/12 part of the shape sphere (Kuwabara & Tanikawa 2010; Moeckel 1989; Moeckel et al. 2012), since the remaining 11/12 are obtained as copies.

In the free-fall three-body problem, we saw that binary collision curves, i.e., curves formed with initial points of orbits which experience binary collision, or simply collision curves, constitute the boundaries of regions in the initial condition plane (Tanikawa & Umehara 1998). We introduced in Tanikawa and Mikkola (2008) symbols so that collision curves form boundaries of regions with different symbol sequences.

In the present report, after the definition of symbols, we start numerical integrations of the free-fall three-body problem. We describe the overall structure of the initial condition plane divided by regions of different symbol

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sequences. We confirm that the boundaries are collision curves. Then, we turn our attention to the structure of two of the three boundaries in the initial condition plane: the circular boundary and the bottom boundary. On these boundaries we successfully look for triple collision orbits and periodic orbits with the help of symbol sequences given to the orbits inside the initial condition plane.

1.1 Equations of motion

We use the logarithmic Hamiltonian method (Mikkola & Tanikawa 2013a, 2013b) for the numerical integration. Let us briefly formulate the equations of motion.

We consider a three-body problem. Let m_i be the mass of the ith body with position $\mathbf{r}_i$ for $i = 1, 2, 3$. We introduce three interparticle distance vectors

$$\boldsymbol{R}_1 = \boldsymbol{r}_3 - \boldsymbol{r}_2, \quad \boldsymbol{R}_2 = \boldsymbol{r}_1 - \boldsymbol{r}_3, \quad \boldsymbol{R}_3 = \boldsymbol{r}_2 - \boldsymbol{r}_1 \tag{1}$$

as new coordinates. Let the corresponding velocities be $\boldsymbol{V}_k = \dot{\boldsymbol{R}}_k$; then the potential and kinetic energies can be written as

$$\begin{aligned} U &= \frac{m_2 m_3}{R_1} + \frac{m_3 m_1}{R_2} + \frac{m_1 m_2}{R_3}, \\ T &= \frac{1}{2M}\left(m_2 m_3 V_1^2 + m_3 m_1 V_2^2 + m_1 m_2 V_3^2\right), \end{aligned} \tag{2}$$

where $R_i = |\boldsymbol{R}_i|$, $V_i = |\boldsymbol{V}_i|$, and $M = \sum_k m_k$.

The equations of motion are

$$\dot{\boldsymbol{R}}_k = \boldsymbol{V}_k, \quad \dot{\boldsymbol{V}}_k = -M\frac{\boldsymbol{R}_k}{R_k^3} + m_k \sum_\nu \frac{\boldsymbol{R}_\nu}{R_\nu^3}. \tag{3}$$

Instead of solving the above equations of motion, we form the equivalent logarithmic Hamiltonian

$$\Lambda = \log\left[T(\boldsymbol{p}) + B\right] - \log\left[U(\boldsymbol{q}, t)\right], \tag{4}$$

where $\boldsymbol{p} = (\boldsymbol{V}_1, \boldsymbol{V}_2, \boldsymbol{V}_3)$, $\boldsymbol{q} = (\boldsymbol{R}_1, \boldsymbol{R}_2, \boldsymbol{R}_3)$, and

$$B = U - T \tag{5}$$

is the negative value of the original Hamiltonian. Defining the new time variable s by

$$dt = ds/(T + B) = ds/U, \tag{6}$$

we obtain equations of motion from the logarithmic Hamiltonian as

$$t' = 1/(T + B), \quad \boldsymbol{R}_k' = \boldsymbol{V}_k/(T + B), \quad \boldsymbol{V}_k' = \dot{\boldsymbol{V}}_k/U, \tag{7}$$

where a prime denotes the derivative with respect to s.

These equations of motion are approximately solved using the leapfrog algorithm. Putting $\boldsymbol{X} = (t, \boldsymbol{q})$ and $\boldsymbol{W} = (B, \boldsymbol{p})$, we can write a single leapfrog step as

$$\begin{aligned} \boldsymbol{X}_{\frac{1}{2}} &= \boldsymbol{X}_0 + \tfrac{1}{2}h\boldsymbol{X}'(\boldsymbol{W}_0), \\ \boldsymbol{W}_1 &= \boldsymbol{W}_0 + h\boldsymbol{W}'_{\frac{1}{2}}, \\ \boldsymbol{X}_1 &= \boldsymbol{X}_{\frac{1}{2}} + \tfrac{1}{2}h\boldsymbol{X}'(\boldsymbol{W}_1), \end{aligned} \tag{8}$$

where h is the step size, and indexes 0 and 1 mean values in the beginning and end of the step. Highly accurate results can be obtained using the above regular leapfrog for the substeps in an extrapolation method, e.g., the Bulirsch–Stoer method (Bulirsch & Stoer 1966).

1.2 Orbits and symbol sequences

In this section we introduce the symbols and symbol sequences, using which preliminary results were announced in Tanikawa and Mikkola (2008).

Considering the extension to three dimensions or to the four-body problem, we use the notion of a simplex and subsimplex. We give a symbol to an orbit when the 2-simplex (triangle) formed with three bodies degenerates into a 1-sub-simplex (an edge of the triangle). More precisely, three bodies generally form a 2-simplex. We give this simplex the usual orientation. Thus, the simplex is positively oriented as denoted by + if three bodies 1, 2, and 3 are arranged in the counter-clockwise order, otherwise the simplex is negatively oriented as denoted by −. For the moment, we consider the planar problem, so the orientation of a 2-simplex in the fixed plane is uniquely determined. Now, suppose that along the orbit the 2-simplex changes sign from + to − at some instant. Then symbol "1" is given to the event, which is usually referred to as the syzygy crossing, if the longest edge is between bodies 2 and 3. Similarly, symbol "2" is given if the longest edge is between bodies 3 and 1, and symbol "3" if the longest edge is between bodies 1 and 2. When the orientation changes sign from − to +, symbol "4" is given to the event if the longest edge is between bodies 2 and 3. Similarly, symbol "5" is given if the longest edge is between bodies 3 and 1, and symbol "6" if the longest edge is between bodies 1 and 2.

Now, let us introduce our three-body problem. This is the so-called free-fall three-body problem (Agekyan & Anosova 1967; Tanikawa et al. 1995; Tanikawa & Mikkola 2008). The geometry of the problem is illustrated in figure 1a. Initially, body m_2 stands still at $(-0.5, 0)$, and body m_3 stands still at $(0.5, 0)$ of the (x, y)-plane. Body m_1 stands still at any position in

$$D = \{(x, y) : x \geq 0, y \geq 0, (x + 0.5)^2 + y^2 \leq 1\}. \tag{9}$$

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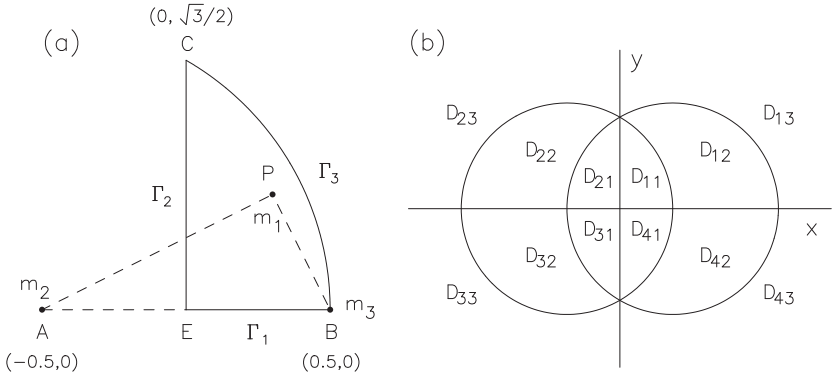


Fig. 1. Geometry of the free-fall problem. (a) We put body m_2 at point A(−0.5, 0), put body m_3 at point B(0.5, 0), and put body m_1 at any point P in the region bounded by segments Γ_1, Γ_2, and circular segment Γ_3 where $\Gamma_1 = \overline{BE}$, $\Gamma_2 = \overline{CE}$, and $\Gamma_3 = \overline{BC}$; coordinates are $E(0, 0)$ and $C(0, \sqrt{3}/2)$. The region bounded by Γ_1, Γ_2, and Γ_3 is sometimes called Anosova's D region. (b) The shape space of the problem. If we paste regions D_{12}, D_{23}, D_{33}, and D_{43} at infinity, the space is changed to the shape sphere. In our case (i.e., the equal mass and zero initial velocity case), 12 regions D_{ij} are all equivalent with each other. Anosova's D region is equivalent with any of them, say, region D_{11}.

The three bodies initially have zero velocities. We further take $m_1 = m_2 = m_3$. Then, the region D of the initial conditions exhausts the form of triangles in which edge $\overline{m_2m_3}$ is the longest, edge $\overline{m_1m_2}$ is the second longest, and edge $\overline{m_1m_3}$ is the shortest. This is sometimes called Anosova's D-region. This is a part of the shape space (Chenciner & Montgomery 2000; Kuwabara & Tanikawa 2010).

The shape space is expressed either as the shape plane or the shape sphere. The shape plane is depicted in figure 1b. The shape sphere is obtained by gluing regions D_{13}, D_{23}, D_{33}, and D_{43} at infinity. Then, the x-axis becomes the equator representing the collinear configurations, while the y-axis becomes one of the meridians, representing the isosceles configurations. The intersections of two circles are the north and south poles, representing the Lagrange equilateral configuration. Three Euler collinear configurations sit 120° apart on the equator. In our free-fall case, Anosova's D region, the initial condition space, is any of the twelve regions of the shape sphere, say, D_{11}.

It is necessary to assure that our symbols are well defined. First, symbols {1, 2, 3} and symbols {4, 5, 6} appear alternately. In fact, the state of a 1-sub-simplex (collinear configuration) is not maintained for a finite non-zero time since the force transverse to the longest edge exerted by two bodies is non-zero until the collinear configuration is realized. In other words, the third body necessarily crosses the longest edge since the third body neither stops at this edge nor turns back to the same side of the edge. It is to be noted that the rotating collinear Euler configuration is not allowed in our setting because the total angular momentum is zero. We do not use the symbol sequences on the x-axis.

We regard that binary collisions are regularized, that is, we continue the solutions of the three bodies beyond binary collisions. One needs care in the case of binary collision. In such a case, the configuration becomes collinear. As is well known, the trajectory of a body relative to the other body is a parabola in the vicinity of binary collision. The third body almost stands still compared with the high speed of the binary components. This again shows that the collinear configuration cannot be maintained before and after collision. It is necessary to make an important remark. At binary collision, an orbit may experience collinear configurations not only once, but frequently experiences two consecutive collinear configurations. Thus, two symbols may frequently be given to an orbit at collision. We will confirm this in section 2.

Each time a triple system becomes collinear, a symbol is given according to the rule described at the top of this subsection. Therefore, an orbit, except the one which terminates at triple collision, represented by an infinite continuous curve in the phase space is replaced by a bi-infinite (corresponding to the past and future) symbol sequence. We denote the symbols by s_i and the boundary between the past and future by a period (•), the present state being just after the period. Here we only consider the future symbol sequences. Then a (future) symbol sequence s starting at the present can be written as

$$s = {}_\bullet s_1 s_2 s_2 \ldots \tag{10}$$

Now, suppose that all the points in the initial condition plane have their own symbol sequences (10), that is, the orbits starting at points of the initial condition plane are all integrated to the future. Of course, triple collision orbits have finite symbol sequences. If we truncate the symbol sequences at the nth digit, there are a finite number of possible combinations of symbols in these length-n "words." The point set of the initial condition plane whose symbol

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sequences contain a particular initial word of length n is called an "n-cylinder." For each n, a finite number of n-cylinders divide the initial condition plane into different sections. We ask what kind of points, or equivalently, orbits, constitute the boundaries of these cylinders.

Let us first consider the case $n = 1$, the first digit. It is apparent from figure 1 that initial 2-simplexes have positive orientations, so only "1," "2," or "3" may appear at first. Actually, only "1" appears since m_1 is closest to the center of gravity and, necessarily, m_1 passes through the segment connecting m_2 and m_3. If we increase the length of words, symbol "4," "5," or "6" appears in the even digits, whereas "1," "2," or "3" appears in the odd digits. In some odd digit, take a cylinder whose word has "1" as its last digit. Take a point in this cylinder and suppose that in the corresponding orbit O, body 1 crosses the segment connecting bodies 2 and 3 with finite non-zero distances both from bodies 2 and 3. Then, all neighboring orbits also experience collinear configurations of body 1 between bodies 2 and 3. This means that the starting point of orbit O is inside the set of points of the cylinder. In other words, the boundary points of the cylinder should correspond to collision orbits. We will discuss this point in the following section.

2 Overall structure of the initial condition plane

There are three types of collision curves: A collision curve of type 1 is a curve of initial conditions for which bodies 2 and 3 collide; a collision curve of type 2 is for collision of bodies 3 and 1; and a collision curve of type 3 is for collision of bodies 1 and 2 (Tanikawa et al. 1995; Umehara & Tanikawa 2000; Tanikawa 2000). It is to be noted that the boundaries of the plane, the x-axis, y-axis, and the circular boundary are all collision curves. There are initial points of orbits which end in triple collision. We call these the triple collision points (TCPs) (Tanikawa 2000).

2.1 Collision curves

Let us first consider the global structure of the initial condition plane of the free-fall problem. In Tanikawa and Mikkola (2008), we also considered the non-free-fall case. We concentrate here on the free-fall case. Figures 2 and 3 show the progression of divisions by cylinders of longer words. Numerical integrations of orbits have been done with step 5×10^{-4} for both the x- and y-directions.

At length three of cylinder words, the initial condition plane is for the first time divided into two regions "161" and "162" (see figure 2a). It is to be noted that region "161" means the region of points having symbol

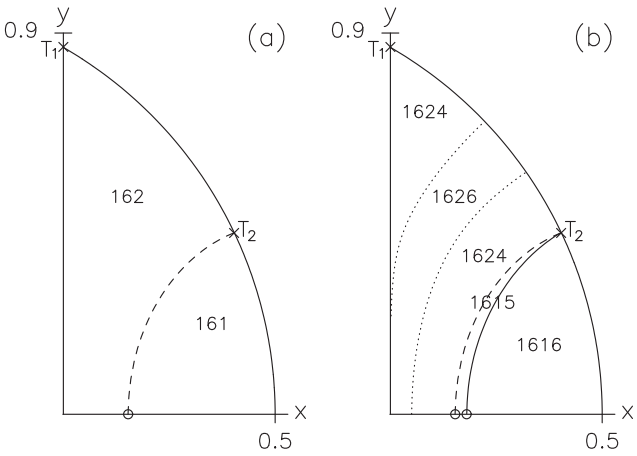


Fig. 2. Structure of the initial condition plane. (a) The first three digits, and (b) the first four digits of symbol sequences. Point T_2 (cross) is one of the TCPs on the circular boundary. Open circles on the x-axis denote TCPs. Fine structures near the y-axis are not shown.

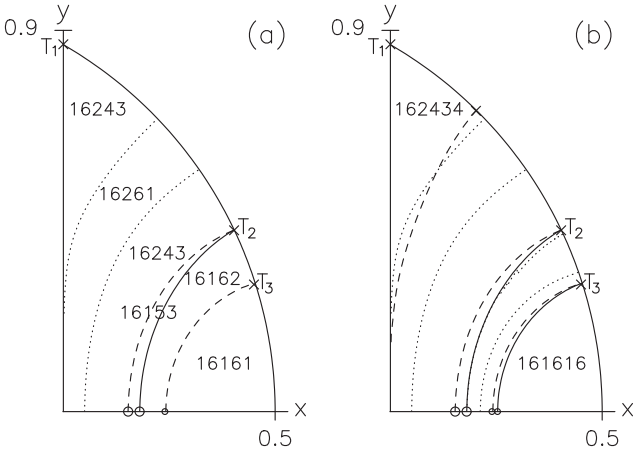


Fig. 3. Overall structure of the initial condition plane. (a) The first five digits, and (b) the first six digits of symbol sequences. Points T_1, T_2, T_3, and open circles on the x-axis denote the TCPs. Fine structures are neglected.

sequences $\bullet 161\ldots$ where $\ldots$ are arbitrary sequences of symbols. Now we ask what is the boundary of these two regions. At the last "1" of "161," body 1 passes through between bodies 2 and 3. At the last "2" of "162," body 2 passes through between bodies 1 and 3. Then at the boundary, bodies 1 and 2 should collide, which implies that the boundary is a collision curve of type 3. One finds that the boundary coincides with a dashed collision curve of type 3 starting at TCP T_2 of figure 8 in Tanikawa, Umehara, and Abe (1995). In the preceding works (e.g., Tanikawa et al. 1995), it was not possible to follow collision curves down to the x-axis due to numerical difficulty. In the present work, it becomes clear that collision curves actually arrive at the x-axis. This is obvious because the two regions "161" and "162" have, in the limit, the x-axis as part of their territories.

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There are clearly general rules: Let s be any word of length greater than 1.

(i) The boundary is a collision curve of type 1 if two regions "s2" and "s3" (resp. "s5" and "s6") are neighboring.

(ii) The boundary is a collision curve of type 2 if two regions "s3" and "s1" (resp. "s6" and "s4") are neighboring.

(iii) The boundary is a collision curve of type 3 if two regions "s1" and "s2" (resp. "s4" and "s5") are neighboring.

In figure 2b, the initial plane is divided by regions corresponding to the 4-cylinders. In principle, possible cylinders are "1614," "1615," "1616," "1624," "1625," and "1626," since "161" and "162" are realized for 3-cylinders. Among these, "1614" and "1625" are not shown in the figure. Actually, region "1614" exists between "1615" and "1616" as a thin region extending from the x-axis up to $y \simeq 0.1$. Region "1625," if it exists, may be between "1624" and "1626." We found substantially no points for this region. We need a finer mesh of integrations to confirm the existence or non-existence of region "1625."

Now, the boundary between regions "1624" and "1626" (a dotted curve) is the collision curve of type 2. For later convenience, we denote this boundary by "162^6_4" with the symbol of the last digit of the left region as subscript and that of the right region as superscript. The boundary "162^4_6" of regions "1626" and "1624" (a dotted curve) is also the collision curve of type 2. The boundary "16^{15}_{24}" is already the boundary in the third digit. This corresponds to the consecutive two syzygy crossings at collision mentioned in the introduction. As a limit of region "162," body 2 passes through bodies 1 and 3. Immediately after this crossing, body 3 passes through between bodies 1 and 2 in a state of collision. As a limit of region "161," body 1 passes through between bodies 2 and 3. Immediately after this crossing, body 2 passes through between bodies 1 and 3 in a state of collision. Finally, the boundary "161^6_5" (a solid curve) is new. It is the collision curve of type 1.

For longer symbol words, the features of the initial plane gradually look like those of figure 2 of Tanikawa and Umehara (1998). A remarkable feature in figure 3b is that collision curves cross inside the initial condition plane. The crossing points of different types of collision curves are triple collision points (TCPs) (see, e.g., Tanikawa & Umehara 1998). The present work shows that TCPs are numerically easily obtained.

There is another overall feature of the initial condition plane already obtained in figure 8 of Tanikawa, Umehara, and Abe (1995). We can give the explicit expressions of these collision curves by symbol words. The first four curves have expressions "16^1_2" (dashed) in figure 2a, "161^6_5" (solid), "162^6_4" (dotted), and "162^4_6" (dotted) in figure 2b. The second four curves have expressions "$(16)^2 1^1_2$," "$(16)^2 1^6_5$," "$(16)^2 2^6_4$," and "$(16)^2 2^4_6$," where "$(16)^n$" means the n times repetition of 16. We use this kind of obvious notation from now on. The sequence of these four curves "$(16)^n 1^1_2$," "$(16)^n 1^6_5$," "$(16)^n 2^6_4$," and "$(16)^n 2^4_6$" as $n \to \infty$ accumulate at the lower-right corner of the initial condition plane.

There is an infinite sequence of TCPs T_n, $n = 1, 2, \ldots$ on the circular boundary of the initial condition plane (Tanikawa 2000), where T_1 is the triangular Lagrange point (figure 3). T_n for $n \geq 2$ are obtained as crossing points of "$(16)^{n-1} 1^1_2$" and "$(16)^{n-1} 1^6_5$."

2.2 Collision curves as sections of stable and unstable manifolds of binary collision manifolds

Using the blow-up method of McGehee, we replace binary collision by an invariant manifold, i.e., the torus, over which the flow of the three-body problem extends smoothly. Collision orbits form stable and unstable manifolds of this collision manifold. These manifolds are cylinders flowing out of or flowing in to the top or bottom circle of the torus (Llibre 1982). So, in general, we get a circular curve if we cut the manifold with a two-dimensional surface. This is in fact observed in our case: see figures 2 and 3. As we already pointed out, Anosova's D-region is 1/12 of the shape sphere. Due to the symmetry coming from zero initial velocities and equal masses, the remaining 11/12 can be obtained by reflecting the D-region with respect to its boundaries.

Now, let us take as an example collision curve 16^1_2 in figure 2a. Let us identify Anosova's D-region with D_{11} in figure 1b. This curve has copies in D_{12}, D_{41}, and D_{42}. Together with the three copies, this curve forms a closed curve encircling the common boundary point of these regions (i.e., point B of figure 1a). This closed curve itself has copies on the shape sphere. One passes through D_{21}, D_{22}, D_{32}, and D_{31}, while the other passes through D_{13}, D_{23}, D_{33}, and D_{43}. There is an infinite number of closed collision curves encircling point B as figures 2 and 3 suggest.

In our case, a circular binary collision curve is not only the section of the stable manifold, but also of the unstable manifold of the binary collision manifold. Take any point on this circle. Then, the orbit starting at this point goes to binary collision in the future. Since the initial velocities are zero, the backward orbit traces the same trajectory. So, the orbit goes to the same binary collision, i.e., the orbit tends to the same binary collision manifold in the past. Thus, each point of a collision curve is a homoclinic point. We may call the curve a *homoclinic curve*.

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Moeckel (1989) showed the connections among different triple collisions using the facts that the equilibrium points (triple collisions) have different heights on the triple collision manifold on which the flow is gradient-like. We can show the existence of connections using binary collision curves. In fact, we see that each of the collision curves of type 1, i.e., the boundary curve $(16)^n 1_5^6$ ($n = 1, 2, \dots$), connects a TCP T_n on the circular boundary and a TCP on the x-axis. Thus, we can say that the stable and unstable manifolds to the two-body collision manifold connect the stable and/or unstable manifolds of different triple collision manifolds. For example, the one-dimensional triple collision manifold and the isosceles triple collision manifold are connected by the stable and unstable manifolds of the two-body collision manifold. This implies that there are orbits which start in any small neighborhood of a TCP T_n (the equilateral triple collision) and pass through any small neighborhood of a TCP on the x-axis (one of the collinear triple collisions), and vice versa.

The stable manifold of the two-body collision manifold entering the one-dimensional triple collision manifold is unique, whereas there are many manifolds entering the isosceles triple manifold. The latter phenomenon is seen in figures 6 and 7 of Tanikawa (2000) in which more than one collision curves enter T_2. This means that the equilateral triple collision has connections to various types of triple collision by stable and unstable manifolds of binary collision manifold.

3 Orbits on the circular boundary and the *x*-axis

We are interested in the nature of orbits of the terminal points of the collision curves at the boundaries of the initial condition plane. We are also interested in the periodic orbits on these boundaries. We already know that TCPs T_i ($i = 1, 2, \dots$) are on the circular boundary (Tanikawa & Umehara 1998). We expect that there is a sequence of TCPs on the x-axis. We know that collision curves passes through these points in the shape sphere.

3.1 Orbits on the circular boundary

We look for the initial coordinates (x, y) of the free-fall periodic orbits on the circular boundary. In this case, the problem is isosceles with base masses m_1 and m_3 (figure 1a). Simo and Martinez (1988) studied the isosceles periodic orbits, among which the free-fall periodic orbits are included; see, in particular, Propositions 12.1, 12.2 and related figures 12.1(a), 12.1(c), and 12.2(a) of their paper. Moeckel, Montgomery, and Venturelli (2012) studied the

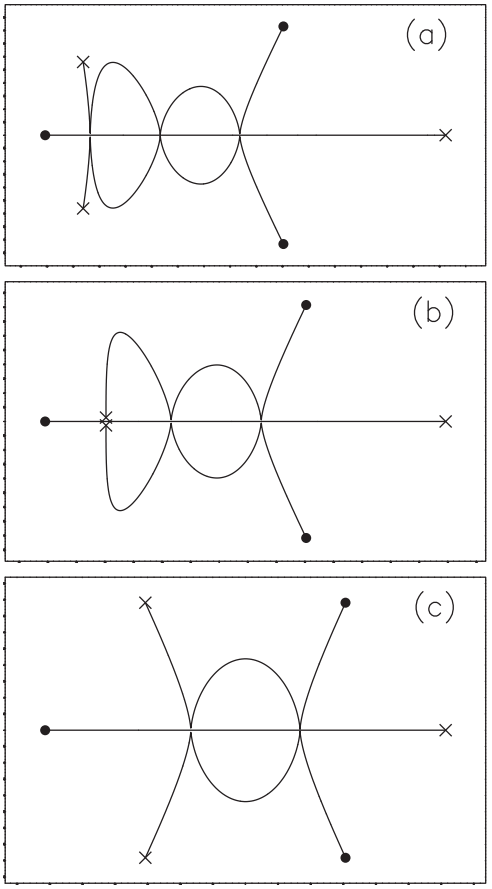


Fig. 4. Looking for periodic orbits on the circular boundary. (a) $x = 0.1546$, $y = 0.7560$ (denoted by FF-N2 in which FF means free-falls at both terminals, N means that the orbit is non-symmetric, and 2 means that the orbit has two loops). (b) $x = 0.1677$, $y = 0.7444$ (denoted by FB-N2 in which FB means a free-fall at one terminal, while a binary collision at the other terminal). (c) $x = 0.1945$, $y = 0.7195$ (denoted by FF-S1 in which S means that the orbit is symmetric). In the three orbits, trajectories start at positions •, and arrive in half the period at positions ×. All orbits trace back the same routes. These orbits are between T_1 and T_2.

brake (i.e., free-fall) periodic orbits with the aid of the variational method. Their result is summarized in Theorem 2 of their paper. Our contribution in this report for free-fall periodic orbits is to position these orbits in the initial condition space.

In order that an orbit be periodic, it must satisfy certain symmetry conditions. This is clearly seen in figure 4 where three types of periodic orbits are shown. These orbits are the free-fall periodic orbits. These orbits at half the period either come to the other free-fall state (figure 4a) or to the symmetric collision condition (figure 4b) in which the third body not involved in collision stands still, and the collision velocity of the binary is perpendicular to the line connecting the binary and the third body. The third orbit (figure 4c) is mirror symmetric with respect to the center of the orbit at the one-fourth period. The first orbit will be called the

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Table 1. Example for the circular boundary.*

Name	x	t	x	t
FF-N2	0.1546	1.7234	0.42154	1.0261
FB-N2	0.1677	1.5184	—	—
FF-S1	0.1945	0.7195	—	—
T_2	0.40351	0.6467	—	—
FF-S3	0.417512	1.2923	—	—
FB-N3	0.41879	1.1694	—	—
FF-N2	0.42154	1.0261	0.1546	1.7234
T_3	0.45298	0.6448	—	—
FF-N4	0.45754	1.1249	0.416192	1.5192
FB-N4	0.457956	1.0321	—	—
FF-N3	0.4588625	0.9259	0.14073	2.1014
T_4	0.47032	0.6452	—	—
FF-N5	0.472617	1.0279	0.415475	1.7323
FB-N5	0.4728136	0.9553	—	—
FF-N4	0.4732438	0.8717	0.133070	2.4574
T_5	0.47899	0.6446	—	—

*The x-coordinates of isosceles periodic orbits and TCPs T_n on the circular boundary. t denotes one half of the period.

FF type, and is denoted by FF-N2 which means starting at a free-fall state and ending at a free-fall state without symmetry and with two loops. The second orbit will be called the FB type, and is denoted by FB-N2 which means starting at a free-fall state and ending at binary collision with two loops. The third orbit is called the symmetric FF type, an is denoted by FF-S1 which means starting at a free-fall state and ending at a free-fall state with symmetry and with one loop.

The orbits of the terminal points of type-2 curves do not in general satisfy these conditions for periodic orbits. Thus, we looked for the free-fall periodic orbits in other places and actually found such orbits. Table 1 gives the example results of the search for the circular boundary. The table also gives the positions of TCPs T_i. In the table, the first column gives the name of the orbit, the second column the x-position on the circle, and the third column the half-period. The fourth and fifth columns give the position and half-period of the orbit whose initial condition is the final condition of the orbit of the second and third columns. The half-periods are different because the two orbits are scaled differently.

3.2 Sequences of orbits ending in triple collision on the x-axis

The position of a TCP on the x-axis can be obtained as accurately as possible by interpolation. Let us explain the procedure of interpolation using the orbits shown in figure 5. There are three panels and in each panel trajectories of the three bodies are plotted. The ordinates are the x-coordinates

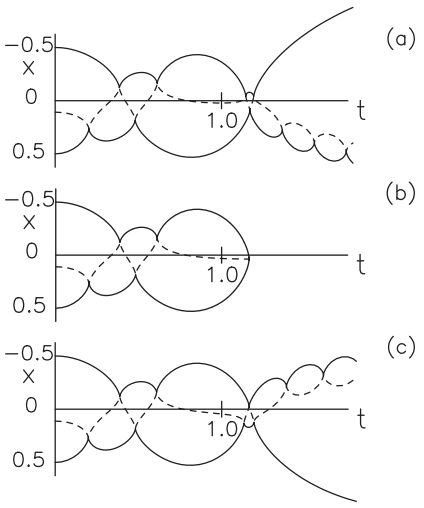


Fig. 5. Looking for a triple collision orbit. (a) $x = 0.10800\ldots$, (b) $x = 0.10968\ldots$, (c) $x = 0.11200\ldots$.

of the three bodies. The abscissa represents the time. In each figure, the top solid curve represents the trajectory of m_2 starting at $(-0.5, 0)$, the middle dashed curve that of m_1 starting at $(x, 0)$ with $0 \leq x < 0.5$, and the bottom solid curve that of m_3 starting at $(0.5, 0)$. Figure 5a shows an orbit starting at a slightly smaller x-coordinate of m_1. In this case, three bodies make a close triple encounter at $t = 1.165\ldots$, then m_1 and m_3 form a binary, and m_2 escapes. Figure 5c shows an orbit starting at a slightly larger x-coordinate. In this case, m_1 and m_2 form a binary after the close triple encounter, and m_3 escapes. A TCO (triple collision orbit) exists in between. In fact, figure 5b shows the orbit of triple collision at the fifth binary collision.

Now, let us present an important sequence of TCPs accumulating at the lower-right corner of the initial condition plane. The sequence has clear geometrical characteristics (see figure 6). We can characterize the form of orbits by the number of lobes which are bounded by trajectories of the bodies. As an example, figure 6a shows a TCO starting at $x = 0$, that is, body 1 starts in the middle of bodies 2 and 3. Three bodies collide at the first collapse. In this case, there are two lobes. Figures 6b, 6c, and 6d show the TCOs at the second, third, and fourth binary collisions of bodies 1 and 3 corresponding to the numbers of lobes 3, 4, and 5. Table 2 gives the further result. The first column shows the starting x-position of the TCPs, the second column the number of lobes, and the third and fourth columns the number n of boundary curves "$(16)^n 1_5^6$" and their numerical positions. This confirms that terminals at the x-axis of curves "$(16)^n 1_5^6$" are TCPs.

There is an infinite number of sequences of TCPs. Let us explain the reason using the orbits in figure 5. The immediate left and right positions (in the x-coordinate) of triple

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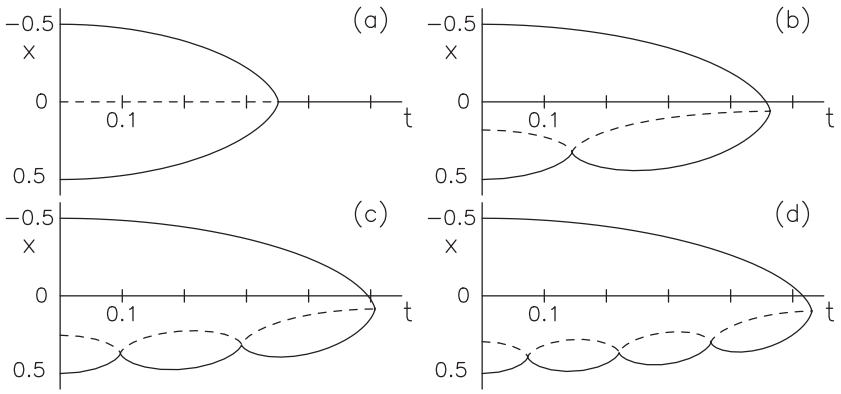


Fig. 6. Sequence of triple collision orbits. (a) $x = 0$, (b) $x = 0.18059\ldots$, (c) $x = 0.25326\ldots$, and (d) $x = 0.29461\ldots$.

Table 2. The sequence of TCPs accumulating at the lower-right corner of the initial condition plane.

x-position of TCPs	No. lobes	Boundary x-position of words $(16)^n 1^6_5$	
		n	Position
0	2		
0.18059...	3	1	0.18061
0.25326...	4	2	0.25328
0.29461...	5	3	0.29463
0.32192...	6	4	0.32193
0.34156...	7	5	0.34157
0.356507...	8	6	0.35651
0.368327...	9	7	0.36833
0.377958...	10	8	0.37796
0.385986...	11	9	0.38598
0.392801...	12	10	0.39280
0.398674...	13	11	0.39868
0.403796...	14	12	0.40380
0.408312...	15	13	0.40833
0.412329...	16	14	0.41235
0.415930...	17	15	0.41593

collision are occupied by "escape orbits" (McGehee 1974) as numerically seen in the top and bottom orbits of figure 5. As the starting position gradually becomes farther from the position of a TCO, orbits cease to escape. There is a critical distance at which the orbit escapes in a parabolic manner. Until this orbit, the number of lobes formed by the binary is infinite. As the distance further increases, the number of lobes formed by the binary after the final triple encounter decreases. Each time the number of lobes changes, a TCO appears. Thus, in the neighborhood of the TCP of figure 5, there are two infinite sequences of TCPs. We initially have an infinite sequence of TCPs as in table 2. Two infinite sequences of TCPs accompany each of the TCPs in table 2. In turn, two new infinite sequences of TCPs accompany TCPs already obtained. This continues indefinitely.

Table 3. Other two sequences of TCPs with orbits $x = 0.10968\ldots$ and $x = 18059\ldots$ as mothers on the x-axis.

x-position	No. lobes	x-position	No. lobes
0	2	0.15660...	7
...		0.16704...	8
0.008739...	16	0.170957...	9
0.009398...	15	0.172943...	10
0.010232...	14	0.174131...	11
0.011321...	13	0.174918...	12
0.012798...	12	0.1754783...	13
0.014908...	11	0.175896...	14
0.01814...	10	0.176221...	15
0.02360...	9	0.176480...	16
0.03418...	8	0.176692...	17
0.05817...	7	...	
0.10968...	6	0.18059...	3

As examples, we give in table 3 a sequence of TCPs which accompanies the TCP at $x = 0$, and a sequence which accompanies the TCP at $x = 0.18059$. The first four trajectories of the former sequence are shown in figure 7. As we explained above, the number of lobes decreases when the distance from the original triple collision increases.

3.3 Sequences of periodic orbits on the x-axis

Let us explain using figure 8 how to find periodic orbits on the x-axis. The vertical dashed segments of the left panels in figure 8 show the instants when bodies come to a standstill. In panels (a) and (c), different bodies come to a standstill at different times. In panel (b), three bodies come to a standstill at the same moment. Then, this orbit is periodic since there are two instants when three bodies come to a standstill. The second instant appears at half the period. Now, the vertical dashed segments of the right panels in figure 8 show the instants with binary collisions, and the third body comes

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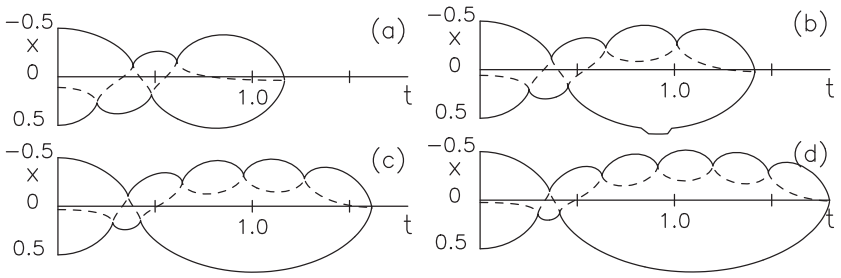


Fig. 7. Sequence of triple collision orbits. (a) $x = 0.10968\ldots$, (b) $x = 0.05817\ldots$, (c) $x = 0.03418\ldots$, and (d) $x = 0.02360\ldots$.

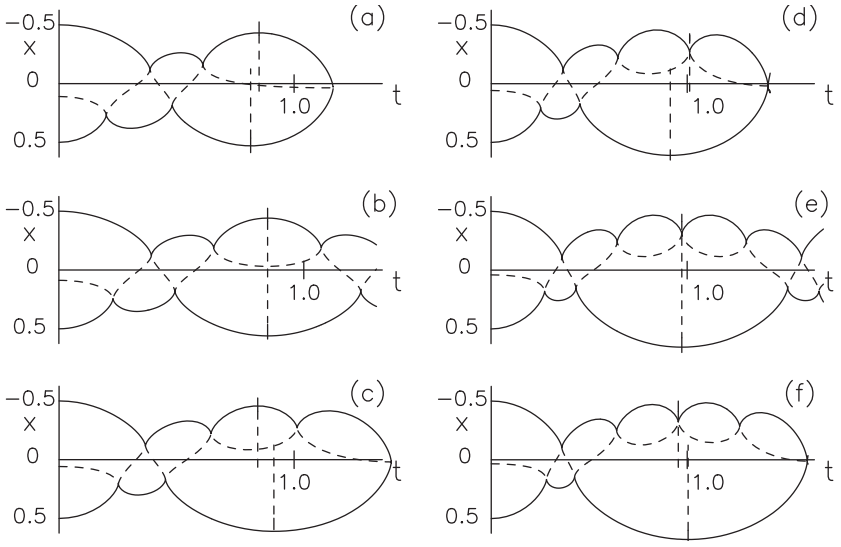


Fig. 8. Left panels: Looking for a periodic orbit of FF type. (a) $x = 0.10968\ldots$, (b) $x = 0.08800\ldots$, (c) $x = 0.05817\ldots$. Right panels: Looking for a periodic orbit of FB type. (d) $x = 0.05817\ldots$, (e) $x = 0.040495$, (f) $x = 0.03418\ldots$.

to a standstill. In panels (d) and (f), the third body stands still before and after the binary collision between bodies 1 and 2. In panel (e), two phenomena occur simultaneously. Then, this orbit is periodic since the standstill and binary collision are time reversible.

Similarly to the case of triple collision orbits, there is an infinite number of sequences of periodic orbits. The argument using figure 5 applies to periodic orbits. In fact, along with the decrease of lobes in the ejection orbits, periodic orbits either of FF-type or FB-type are shown to exist. We omit the numerical results for these orbits.

4 Concluding remarks

We have demonstrated that the use of symbol sequences is effective in the planar three-body problem. The main results are:

(i) In the free-fall problem, it has been shown that the stable and/or unstable manifolds of two-body collision manifolds (in other words, binary collision curves) divide the initial condition plane. This has been suggested in Tanikawa, Umehara, and Abe (1995). In the present work, divisions are done by symbol sequences and more clearly seen.

(ii) We have numerically shown the smooth connections of TCPs on the circular boundary and the x-axis by the stable and/or unstable manifolds of two-body collision manifolds. In addition, these stable and unstable manifolds form circles on the shape sphere.

(iii) We have positioned the sequence of isosceles free-fall periodic orbits on the circular boundary of Anosova's D-region. We have numerically shown the existence of an infinite number of sequences of TCPs and periodic points along the axis, i.e., in the free-fall collinear three-body problem.

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